

WASP-11b

R-1265

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_150104 · created 2026-07-31T13:09:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457026.6791 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.011
Transit depth (Rp/Rs)^2	2.0 +/- 0.32 [%]
Orbital Inclination (inc)	87.7 +/- 0.61
Airmass coefficient 1 (a1)	1.193 +/- 0.0081
Airmass coefficient 2 (a2)	-0.156 +/- 0.026
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	None
Optimal Aperture	3.83
Optimal Annulus	16.99
Transit Duration (day)	0.0771 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.011 vs 0.131 (deviation 0.62 σ)
Duration fit vs published	0.0771 d vs 0.1075 d (deviation 6.74 σ)
Mid-time O-C	2.7 min
Depth SNR	8.8
Residual scatter	0.68 %

Light curve

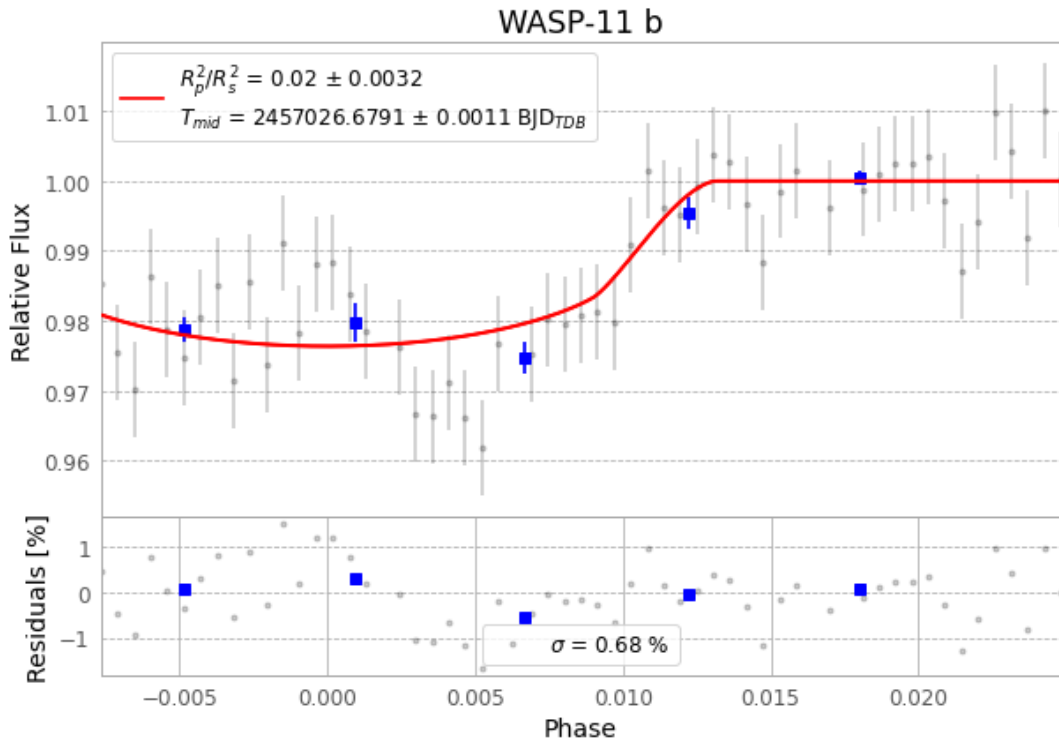


Figure 1 — FinalLightCurve_WASP-11 b_2015-01-03.png (SHA-256 250012b02a4832bb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 272], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aee973160fe9c6c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), HATP-10150104033012.FITS → HATP-10150104062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-150104.log (f3e72695e138...), FinalLightCurve_WASP-11 b_2015-01-03.png (250012b02a48...), FinalLightCurve_WASP-11 b_2015-01-03.csv (3fa5a016ce71...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 13:09Z.